

SAMSUN, Turkey

WMO: 170300

Lat: 41.3439N

Lon: 36.2562E

Elev: 13

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.4	33.7	16.5	12.7	41.4	20.0	15.0	42.3	20.5	50.6	18.4	52.2	5.4	290	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	84.6	72.5	83.1	72.2	81.8	71.6	75.3	82.1	74.2	81.2	73.3	80.3	6.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	121.9	80.4	71.7	117.3	79.8	70.6	112.7	79.3	38.8	82.5	37.8	81.3	36.9	80.3	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.4	10.6	DB	29.2	92.3	3.6	3.0	26.6	94.4	24.5	96.2	22.4	97.8	19.8	100.0	(4)
(5)			WB	26.0	77.6	3.9	1.7	23.2	78.8	21.0	79.8	18.8	80.8	15.9	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	46.0	46.7	48.4	53.6	61.6	69.7	74.4	75.9	69.6	62.2	55.1	49.3	(6)	
	DBStd	11.98	6.63	7.49	6.60	6.24	5.13	4.21	2.79	3.21	4.60	5.45	6.64	6.90	(7)	
	HDD50	562	160	144	112	27	0	0	0	0	0	0	23	96	(8)	
	HDD65	3031	588	513	514	349	131	9	0	13	123	304	487		(9)	
	CDD50	4017	37	51	64	134	360	592	757	802	589	378	178	75	(10)	
	CDD65	1009	0	1	1	6	25	151	292	337	152	35	8	1	(11)	
	CDH74	5844	0	2	12	58	68	567	1865	2482	664	94	32	0	(12)	
	CDH80	797	0	0	1	19	12	48	208	431	56	20	2	0	(13)	
(14) Wind	WSAvg	3.9	5.0	4.5	3.7	3.2	3.0	3.6	4.4	4.2	3.9	3.3	3.7	4.8	(14)	
(15) Precipitation	PrecAvg	27.7	2.6	2.0	2.4	2.3	2.0	1.8	1.2	1.5	2.0	3.2	3.3	3.2	(15)	
	PrecMax	39.8	6.1	5.0	5.6	5.7	7.4	5.9	4.6	6.5	5.2	10.1	13.8	9.4	(16)	
	PrecMin	19.8	0.2	0.6	0.2	0.3	0.3	0.1	0.0	0.0	0.2	0.2	0.2	0.7	(17)	
	PrecStd	4.3	1.3	1.0	1.1	1.2	1.3	1.1	1.0	1.5	1.3	2.1	2.2	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.6	72.1	75.0	80.9	80.9	83.9	85.4	87.8	84.5	82.1	78.0	69.7	(19)	
		MCWB	51.1	52.8	55.2	59.9	65.1	71.9	73.0	73.8	69.1	63.8	56.5	53.2	(20)	
	2%	DB	63.2	67.1	67.9	72.6	74.7	81.0	83.6	85.1	81.5	76.0	72.1	66.1	(21)	
		MCWB	48.7	51.3	52.7	58.5	64.0	71.4	72.7	73.6	69.9	64.8	55.5	51.8	(22)	
	5%	DB	59.4	62.5	62.9	66.6	71.8	79.1	82.3	83.7	79.5	73.0	68.1	62.8	(23)	
		MCWB	47.2	49.3	51.2	56.6	63.3	70.3	72.3	73.1	69.7	64.4	55.1	50.7	(24)	
	10%	DB	55.9	58.4	58.1	62.6	69.3	77.1	80.9	82.4	78.0	70.3	64.9	59.3	(25)	
		MCWB	46.0	47.3	49.5	54.4	62.3	68.8	71.5	72.5	69.1	63.2	54.1	49.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	54.5	56.4	62.0	67.8	75.0	76.2	77.3	73.7	70.9	61.7	55.4	(27)	
		MCDB	62.1	68.1	71.1	75.1	74.9	81.3	81.6	83.9	79.7	75.6	67.4	64.9	(28)	
	2%	WB	50.3	51.9	53.8	59.3	65.6	73.2	74.9	75.6	72.5	67.7	59.1	53.6	(29)	
		MCDB	58.8	64.7	65.9	70.9	72.1	79.4	81.4	82.8	79.0	73.0	65.9	62.1	(30)	
	5%	WB	48.7	50.1	51.9	57.1	64.2	71.5	73.9	74.6	71.3	65.7	57.3	52.1	(31)	
		MCDB	55.7	60.0	60.8	65.1	70.0	78.3	80.7	81.9	78.1	70.8	64.6	59.3	(32)	
	10%	WB	47.4	48.4	50.3	55.3	62.8	69.8	72.8	73.7	69.9	64.0	55.8	50.5	(33)	
		MCDB	53.5	56.4	56.7	61.6	68.5	76.4	79.8	81.0	77.2	69.3	63.3	57.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.9	11.8	11.5	10.7	11.2	11.6	11.6	12.2	11.0	12.2	10.3	(35)	
		MCDBR	15.9	19.0	21.6	20.2	14.2	13.1	12.6	12.3	14.1	15.3	17.8	15.3	(36)	
	5% WB	MCWBR	7.7	8.8	9.9	9.8	7.9	9.5	9.2	8.8	10.4	9.2	9.2	7.5	(37)	
		MCWBR	13.0	16.5	18.8	17.0	12.1	12.4	11.9	12.0	12.9	12.2	13.9	12.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.355	0.391	0.436	0.480	0.450	0.423	0.439	0.449	0.419	0.409	0.368	0.354	(40)	
		taud	2.321	2.214	2.113	2.032	2.202	2.326	2.302	2.264	2.337	2.303	2.365	2.347	(41)	
	Ebn at Noon		246	253	255	252	264	271	265	259	259	245	240	236	(42)	
		Edh at Noon	28	36	45	52	45	40	41	41	36	33	27	26	(43)	
(44) All-Sky Solar Radiation	RadAvg		505	738	1073	1492	1817	2094	2086	1831	1406	928	643	451	(44)	
	RadStd		54	90	82	153	180	166	108	103	99	85	73	55	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	-148	-257	+191	+119	

Nomenclature: See separate page